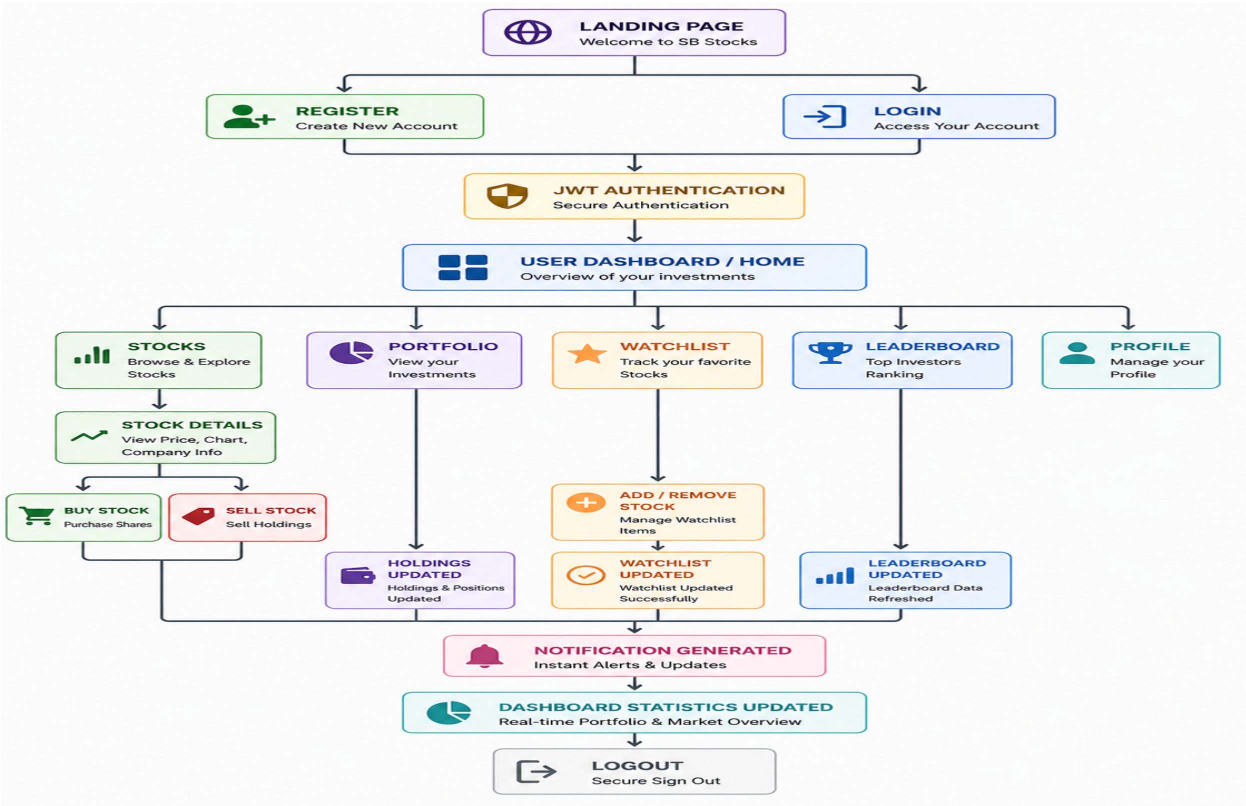
MVC PATTERN



Description

The SB Stocks application follows the Model-View-Controller (MVC) architectural pattern, separating the system into Model, View (Routes), and Controller layers. The User sends requests through the frontend, Routes forward them to the Controller, the Controller processes business logic and communicates with the Model, and the Model interacts with MongoDB using Mongoose before returning responses.

User (Client)

Represents the React.js frontend. Users perform registration, login, stock browsing, buy/sell operations, portfolio management, watchlist updates, notifications, and profile management through REST APIs.

View Layer (Routing Layer)

Implements REST API endpoints including authRoutes.js, stockRoutes.js, portfolioRoutes.js, transactionRoutes.js, watchlistRoutes.js, goalRoutes.js, notificationRoutes.js, leaderboardRoutes.js, and adminRoutes.js. It maps requests to controllers.

Controller Layer

Receives requests, validates input, executes business logic, invokes model methods, handles exceptions, and returns JSON responses.

Model Layer (Data Layer)

Uses Mongoose schemas for User, Stock, Portfolio, Transaction, Watchlist, Notification, Goal, and Leaderboard. Responsible for CRUD operations and validation.

Database Layer

MongoDB stores users, stocks, portfolios, transactions, watchlists, notifications, goals, and leaderboard records.

MVC Request Flow

1. User sends request.
2. Route receives request.
3. Controller validates and processes.
4. Controller calls Model.
5. Model performs CRUD on MongoDB.
6. Database returns data.
7. Controller returns JSON response.

Advantages of Using MVC

- Separation of Concerns
- Scalability
- Maintainability
- Code Reusability
- Data Integrity
- Easy Testing
- Collaboration-Friendly